

ZETU KULUVUYO ZIDE

FOUNDATIONAL WEB DEVELOPMENT

Capstone 2: Debug Task Manager

Due Date: 13 July 2026

Issues identified in Starter Task Management Application Codebase

The following errors and omissions were identified across *app.js*, *dom.js*, *utils.js* and *app.test.js*.

1. Variables and Operators

In *app.js*, the following issues were identified:

- `taskList` is declared without a `let/const/var` which then creates an implicit global variable.
- `taskCounter` is declared with `var`, it should be `let` and not `var`. All local variables declared with `var` must be replaced with either `let` or `const` throughout.
- Assignment operator `'='` is used inside an if-statement instead of strict equality `'==='`.
- Loose equality `'=='` is used in while loop, it should be `'==='`.

In *utils.js*, the following issues were identified:

- Loose equality `'=='` is used to compare task priority, it should be `'==='`.
- `isHighPriority` returns a string `'yes'` or `'no'` instead of a Boolean which will return either a `true` or `false`.
- `Priorities` array is declared with `var`, it should be `const`.

2. Control Flow

In *app.js*, the following issues were identified:

- The for loop uses `i <= taskList.Length` which is off by one, thereby causing undefined access on the last iteration since the index will be out of bounds.
- The while loop has no `i++` increment which then results in an infinite loop. (line 60)
- All loops use old style indexing, should be replaced with for-of loops, a minimum of 2 is required as per checklist.
- `countCompletedTasks` recursive function is missing a base case and therefore will always throw a `RangeError`.
- `calculateAveragePriority` has no guard for empty `taskList` which will then return `NaN` when the array is empty.

In *dom.js*, the following issues were identified:

- `setupEventListeners()` is called directly at the bottom of the file which runs it before the DOM is ready, it should be inside `DOMContentLoaded`.

3. Functions and Functional Programming

In *app.js*, the following issues were identified:

- `findTaskByTitle()` is missing its `title` parameter.
- `countCompletedTasks` has no base case check and so the recursion never terminates.
- No array methods used anywhere i.e. `map`, `filter`, `reduce`, `find` etc. a minimum of 5 is required as per checklist.
- No higher-order functions implemented.
- All functions reference or modify global `taskList`.

- mergeTasks uses manual loops instead of the spread operator.
- getHighPriorityTasks uses manual loop instead of Array.filter.
- No rest parameters have been used anywhere.
- displayAllTasks uses console.log inside a broken loop, this is not a functional approach.

In *utils.js*, the following issues were identified:

- generateRandomId uses Math.random() returning a decimal, should use Math.floor with multiplication for an integer ID.

4. OOP and classes

In *app.js*, the following issues were identified:

- Task constructor missing id property assignment and so each task won't have a unique identifier.
- Task class is missing toggleCompletion() method which is required as per checklist.
- SubTask constructor is missing the super() call entirely
- SubTask constructor assigns this.parentTask before super(), this is invalid in a derived class.
- TaskManager object only has one method, there's still methods missing.
- TaskManager.tasks references taskList at definition time which will not stay in sync after tasks are added.

5. Modern JavaScript / ES6+ Features

- No destructuring used in any function parameters anywhere across all files.
- No ES6 module export/import statements in any file.

In *app.js*, the following issues were identified:

- getInfo() uses string concatenation instead of template literals.
- getTaskDetails manually copies each property instead of using object destructuring.
- mergeTasks uses manual loops instead of the spread operator to combine arrays.

In *dom.js*, the following issues were identified:

- innerHTML built by concatenation instead of template literals.

6. DOM Manipulation

In *dom.js*, the following issues were identified:

- document.getElementById('add-task-btn') mixes ID method with a class selector, should use querySelector('add-task-btn').
- document.querySelector('task-input') missing # for ID selector, should be '#task-input' or match the actual HTML element id.
- No null checks before calling addEventListener or manipulation DOM elements.
- displayTasks() does not clear the container before re-rendering which causes duplicate task entries.
- handleTaskClick uses event.target.id to get task ID, this is incorrect, task ID is stored in a data attribute not element ID.
- No event delegation implemented.

In *utils.js*, the following issues were identified:

- saveToStorage passes raw data to localStorage.setItem without JSON.stringify
- loadFromStorage returns raw string from localStorage without JSON.parse.

7. Testing

In *app.test.js*, the following issues were identified:

- No import/require statement for Task, addTask, or any functions, tests cannot run.
- All test blocks use var instead of const and contain no beforeEach to reset state between tests.
- Only 2 tests are present, a minimum of 10 passing tests requires as per checklist.
- No edge case tests i.e. empty arrays, null/undefined inputs, and invalid priorities are untested.

8. Error Handling and Code Quality

- No try-catch blocks anywhere across all files.
- No parameter validation in any function e.g. addTask accepts undefined title and priority without error.
- Insufficient code comments, this codebase only has inline placeholder comments.